QUANTITATIVE METHODS 1

Worksheet 4 – Bivariate distributions and regression

Exercises

Exercise 1 – Complete linear regression

Consider $(1,3),(2,5),(3,5),(4,8),(5,9)$.

1. Draw the scatter plot and describe direction, form and apparent strength.
2. Calculate $\bar{x}$, $\bar{y}$, $V(x)$, $V(y)$, $\mathrm{Cov}(x,y)$ and correlation r.
3. Estimate the least-squares line $\hat{y} = ax + b$.
4. Calculate fitted values and residuals $e_i = y_i - \hat{y}_i$.
5. Calculate SSE, SST and R^2 and interpret R^2.

Exercise 2 – Interpolation, extrapolation, causality and influence

Using Exercise 1, forecast Y at $X = 3.5$ and $X = 10$, classify each forecast, add point $(10, 30)$ and recalculate the regression line, then explain why high correlation does not establish causality.

Exercise 3 – Mayer line

For $(1,7),(2,10),(3,10),(4,14),(5,14)$, form two sub-clouds, calculate mean points A and B and determine the Mayer line.

Exercise 4 – Two-stage forecast

Month	1	2	3	4	5	6	7	8
Advertising X	12	15	14	18	20	22	25	27
Sales Y	48	52	50	59	63	68	73	78

Given $\widehat{Y} = 2.064X + 21.899$ and $\widehat{X}_t = 2.155t + 9.429$, forecast month-10 advertising and sales, and identify where extrapolation occurs.